THE NOVA RESEARCH CURRICULUM

Answer key · Grade 8 · Term 2

For teachers only. Do not share with students.

Teacher answer key

English

Question bank

#	Outcome	Question	Answer
1	EN-8.2	If I ___ earlier, I would have caught the bus.	left / leave / **had left** / would leave
2	EN-8.2	Which is a third conditional?	If it rains, I stay in. / If it rains, I will stay in. / If it rained, I would stay in. / **If it had rained, I would have stayed in.**
3	EN-8.2	Choose the correct punctuation: "It was late ___ we went home."	, / **;** / : / ?
4	EN-8.2	A colon can introduce a list. True or false.	**True**
5	EN-8.3	A bibliography lists ...	the chapters / **the sources used** / the writer's friends / new ideas
6	EN-8.3	A research report should use headings. True or false.	**True**
7	EN-8.3	Write a reference for a book.	Model: Khan, A. (2020). Rivers of Pakistan. Lahore: Example Press.
8	EN-8.6	Copying a paragraph without credit is ...	research / **plagiarism** / summarising / editing

Urdu

Question bank

#	Outcome	Question	Answer
1	UR-8.2	«اس کا چہرہ چاند کی طرح روشن ہے» میں مشبہ بہ کیا ہے؟	چہرہ / **چاند** / روشن / کی طرح
2	UR-8.2	تشبیہ میں حرفِ تشبیہ ہوتا ہے، استعارے میں نہیں۔ True or false.	**True**
3	UR-8.2	«میدانِ جنگ میں شیر دھاڑ رہا تھا» میں کیا ہے؟	تشبیہ / **استعارہ** / محاورہ / ضرب المثل
4	UR-8.2	تشبیہ کے چار ارکان لکھیں۔	Model: مشبہ، مشبہ بہ، وجہِ شبہ، حرفِ تشبیہ۔

#	Outcome	Question	Answer
5	UR-8.4	اخبار کے نام خط کس کو مخاطب کرتا ہے؟	پرنسپل / دوست / **مدیر** / والد
6	UR-8.4	دلیل والے مضمون میں ہر پیراگراف میں ایک دلیل ہونی چاہیے۔ True or false.	**True**

Mathematics

Question bank

#	Outcome	Question	Answer
1	MA-8.2	Solve x + y = 10 and x − y = 4.	x = 6, y = 4 / **x = 7, y = 3** / x = 5, y = 5 / x = 4, y = 6
2	MA-8.2	In y = 3x − 2, the gradient is …	−2 / **3** / 1 / x
3	MA-8.2	Solve 2x + 1 > 7.	x > 4 / **x > 3** / x < 3 / x = 3
4	MA-8.3	A right-angled triangle has legs 6 cm and 8 cm. Hypotenuse?	14 cm / 12 cm / **10 cm** / 48 cm
5	MA-8.3	Which is a Pythagorean triple?	2, 3, 4 / **5, 12, 13** / 4, 5, 6 / 1, 2, 3
6	MA-8.3	Arc length of a 90° sector with r = 14 cm (π = 22/7)?	44 cm / **22 cm** / 88 cm / 11 cm
7	MA-8.5	Similar shapes always have the same size. True or false.	**False** (same shape; sides in the same ratio)
8	MA-8.5	A 2 cm side enlarged by scale factor 3 becomes …	5 cm / **6 cm** / 2/3 cm / 9 cm
9	MA-8.5	Which rule proves two right-angled triangles congruent using hypotenuse and one side?	SSS / ASA / **RHS** / AAA

Physics

Question bank

#	Outcome	Question	Answer
1	PH-8.2	Current is measured in …	volts / ohms / **amperes** / watts
2	PH-8.2	An ammeter is connected in …	parallel / **series** / either / the plug
3	PH-8.2	V = 12 V, R = 4 Ω. Current?	48 A / **3 A** / 16 A / 8 A
4	PH-8.2	A 230 V heater draws 5 A. Its power?	46 W / 235 W / **1,150 W** / 225 W
5	PH-8.2	A fuse melts when the current is too high. True or false.	**True**
6	PH-8.2	The earth wire protects you when …	the bulb is off / **a metal case becomes live** / the cell is flat / it rains

#	Outcome	Question	Answer
7	PH-8.2	A longer wire has more resistance. True or false.	**True**
8	PH-8.2	A 1,000 W iron is used for 2 hours. How many units (kWh)?	0.5 / 1 / **2** / 2,000

Chemistry

Question bank

#	Outcome	Question	Answer
1	CH-8.2	Balance: __H_2 + O_2 → __H_2O	1 and 1 / **2 and 2** / 2 and 1 / 1 and 2
2	CH-8.2	Zn + $CuSO_4$ → $ZnSO_4$ + Cu is a ... reaction.	decomposition / **displacement** / combustion / neutralisation
3	CH-8.2	A reaction that gives out heat is ...	endothermic / **exothermic** / physical / neutral
4	CH-8.2	Heating calcium carbonate to make calcium oxide and carbon dioxide is decomposition. True or false.	**True**
5	CH-8.3	In an ionic bond, electrons are ...	shared / **transferred** / destroyed / doubled
6	CH-8.3	Which has covalent bonds?	NaCl / MgO / **H_2O** / KCl
7	CH-8.3	Sodium forms an ion with charge ...	−1 / **+1** / +2 / 0
8	CH-8.3	Why do non-metals share electrons?	Model: sharing lets each atom fill its outer shell.

Biology

Question bank

#	Outcome	Question	Answer
1	BI-8.2	Which neuron carries messages to muscles?	sensory / relay / **motor** / receptor
2	BI-8.2	The central nervous system is the brain and ...	heart / **spinal cord** / lungs / nerves in the hand
3	BI-8.2	Which part of the brain controls balance?	cerebrum / **cerebellum** / brain stem / spinal cord
4	BI-8.2	A reflex action is fast and automatic. True or false.	**True**
5	BI-8.2	In a reflex arc, the effector is usually a ...	**muscle** / bone / receptor / neuron

#	Outcome	Question	Answer
6	BI-8.2	Put in order: motor neuron, receptor, relay neuron, sensory neuron.	Model: receptor → sensory neuron → relay neuron → motor neuron.
7	BI-8.2	Why are reflexes useful?	Model: they protect the body from harm quickly, without waiting for conscious thought.
8	BI-8.2	Wearing a helmet helps protect the brain. True or false.	**True**

History, Geography and Pakistan Studies

Question bank

#	Outcome	Question	Answer
1	HI-8.1	Sir Syed founded the M.A.O. College at Aligarh in ...	1857 / **1875** / 1906 / 1920
2	HI-8.1	Asbab-e-Baghawat-e-Hind explains the causes of ...	the Partition / **the revolt of 1857** / the Mughal decline / World War I
3	HI-8.2	The All-India Muslim League was founded in 1906 in ...	Lahore / Delhi / **Dhaka** / Aligarh
4	HI-8.2	Allama Iqbal gave the Allahabad Address in ...	1906 / 1929 / **1930** / 1940
5	HI-8.2	The Quaid presented his Fourteen Points in 1929. True or false.	**True**
6	GE-8.2	Pakistan's largest industry is ...	cars / **textiles** / ships / aircraft
7	GE-8.2	Banking and IT are part of the ... sector.	agriculture / industry / **services** / mining
8	PS-8.1	The minimum voting age in Pakistan is ...	16 / **18** / 21 / 25
9	PS-8.1	Elections in Pakistan are run by the ...	Senate / Supreme Court / **Election Commission of Pakistan** / President

Islamiyat and Ethics

Question bank

#	Outcome	Question	Answer
1	IS-8.1	The qualities of Ibad-ur-Rahman are in Surah ...	Luqman / **Al-Furqan** / Al-Hujurat / Al-Mulk
2	IS-8.1	When the ignorant speak harshly, Ibad-ur-Rahman reply with ...	anger / silence forever / **peace** / insults
3	IS-8.1	In spending they are ...	wasteful / stingy / **balanced** / careless

#	Outcome	Question	Answer
4	IS-8.1	They do not give false witness. True or false.	**True**
5	IS-8.1	Name three qualities of Ibad-ur-Rahman.	Model: they walk humbly, pray at night and are balanced in spending (also: reply with peace, avoid false witness, repent).
6	IS-8.2	The agreement the Prophet ﷺ made with the people of Madinah is the …	Treaty of Hudaybiyyah / **Charter of Madinah** / Farewell Sermon / Pledge of Aqabah
7	IS-8.2	The Prophet ﷺ helped with work at home. True or false.	**True**
8	ET-8.1	If you see cyberbullying, you should …	share it / ignore it always / **report it and support the person** / join in

Computer Science and skills

Question bank

#	Outcome	Question	Answer
1	CS-8.2	Which tag makes the largest heading?	<h6> / **<h1>** / <head> / <p>
2	CS-8.2	Which tag makes a link?	`<link>` / **`<a>`** / `<href>` / `<url>`
3	CS-8.2	The alt text of an image helps people who cannot see it. True or false.	**True**
4	CS-8.2	HTML gives structure; CSS gives …	data / **style** / sound / security
5	CS-8.2	Write the HTML for a bulleted list of two fruits.	Model: `<ul><li>Mango</li> <li>Apple</li></ul>`
6	RO-8.1	After testing a prototype, the next step is to …	stop / **improve it** / sell it / throw it away
7	FL-8.1	Fixed costs Rs 6,000; profit per item Rs 30. Break-even number of items?	20 / **200** / 2,000 / 180
8	FL-8.1	Every business plan should list possible risks. True or false.	**True**